2012 IEEE/ACM 45th Annual International Symposium on Microarchitecture

CoLT: Coalesced Large-Reach TLBs

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Abstract

Translation Lookaside Buffers (TLBs) are critical to system performance, particularly as applications demand larger working sets and with the adoption of virtualization. Architectural support for superpages has previously been proposed to improve TLB performance. By allocating contiguous physical pages to contiguous virtual pages, the operating system (OS) constructs superpages which need just one TLB entry rather than the hundreds required for the constituent base pages. While this greatly reduces TLB misses, these gains are often offset by the implementation difficulties of generating and managing ample contiguity for superpages.

We show, however, that basic OS memory allocation mechanisms such as buddy allocators and memory compaction naturally assign contiguous physical pages to contiguous virtual pages. Our real-system experiments show that while usually insufficient for superpages, these intermediate levels of contiguity exist under various system conditions and even under high load. In response, we propose Coalesced Large-Reach TLBs (CoLT), which leverage this intermediate contiguity to coalesce multiple virtual-to-physical page translations into single TLB entries. We show that CoLT implementations eliminate 40% to 58% of TLB misses on average, improving performance by 14%.

Overall, we demonstrate that the OS naturally generates page allocation contiguity. CoLT exploits this contiguity to eliminate TLB misses for next-generation, big-data applications with low-overhead implementations.

1. Introduction

Translation Lookaside Buffers (TLBs) are crucial to system performance due to their long miss penalties [5, 11, 14, 19, 22, 26]. Past work has shown that TLB misses degrade performance by 5% to 14% for even nominally-sized applications. This number worsens to 50% in virtualized environments or when the application's memory footprint increases [8, 21].

Superpages have previously been proposed to increase TLB coverage [15, 16, 23, 25, 28, 29]. A superpage is a memory page that is sized as a multiple of a base page and is typically in the megabyte or gigabyte range. Each TLB superpage entry can thus replace hundreds of baseline TLB entries, boosting TLB coverage.

By construction, superpages target situations where the OS can seamlessly generate vast amounts of contiguity. Unfortunately, a number of issues may preclude this. For example, superpages may magnify an application's memory footprint, increasing paging traffic. Superpages also require specialized and high-overhead algorithms to assign aligned and contiguous physical page frames to contiguous virtual pages [25, 28, 29]. The OS therefore selectively uses superpages such that the benefits of reduced TLB misses are not outweighed by these overheads.

This work observes that there actually exists a second regime of page allocation contiguity, orthogonal to superpaging. Specifically, OS memory allocators use buddy allocators and memory compaction daemons which also, by construction, allocate contiguous physical page frames to contiguous virtual pages. These mechanisms generate *intermediate contiguity* (in the range of tens of pages), which falls short of superpage requirements (hundreds of contiguous pages). However, this contiguity is achieved without superpaging overheads like increased I/O traffic and sophisticated page construction algorithms. In response, we propose Coalesced Large-Reach TLBs (CoLT), a series of hardware mechanisms that allow TLBs to coalesce multiple contiguous virtual-to-physical address translations, increasing their memory reach. CoLT specifically targets large amounts of intermediate contiguity that superpaging cannot exploit. Our contributions are as follows:

- We study, on a real system, how often consecutive virtual pages are allocated consecutive physical pages. We find that even with high system fragmentation, tens of pages are usually contiguous. Furthermore, while superpaging does increase contiguity, much of it falls short of the level necessary to actually create large pages (a 2MB superpage needs 512 contiguous 4KB base pages, while we see tens of contiguous pages). Instead, we show that TLB coalescing effectively leverages this contiguity.
- We propose CoLT for set-associative, two-level TLB hierarchies commonly found in processors today [3, 17]. Our strategies eliminate roughly 40% of L1 and L2 TLB misses, resulting in average performance improvements of 12%.
- We develop CoLT support for small, fully-associative TLBs commonly found in processors to cache superpage entries [3, 17]. We show how to overcome the challenge of designing these small structures, achieving L1 and L2 TLB miss elimination rates of 58% on average. These translate to average performance improvements of 14%.
- Finally, we combine the benefits of coalescing on both set-associative and fully-associative TLBs, improving average performance by 14%.

Overall, we design low-overhead hardware to exploit intermediate levels of page allocation contiguity. Our studies evaluate this approach under a variety of system configurations with heavy system load and different superpaging configurations.

2. Background and Related Work

2.1. Prior TLB Enhancement Techniques

Address translation, especially with virtualization and larger application working sets, is a primary source of system performance degradation [8, 21]. In response, researchers have considered techniques to improve TLB structure, lookup, and placement [9, 13]. More sophisticated techniques such as TLB prefetching and mechanisms to accelerate page walks have also been considered [5, 11, 19, 27]. Our goal is to propose techniques, orthogonal to past work, to further boost TLB performance.

2.2. Superpaging Benefits and Problems

Superpages or hugepages have previously been proposed to lower TLB miss rates [15, 16, 23, 25, 28, 29]. Superpages are typically

1072-4451/12 $26.00 © 2012 IEEE
DOI 10.1109/MICRO.2012.32

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